

■ INTRODUCTION

The CE6218 Series are a group of positive voltage regulators manufactured by CMOS technologies with high ripple rejection, extremely low power consumption and low dropout voltage, which consume less than 0.1 μ A in shutdown mode and can provide large output currents even when the difference of the input-output voltage is small. Thus the series are very suitable for the battery-powered equipments, such as Portable/Palm computers, Portable consumer equipments, industry equipments and so on, which want to prolong the using life of the battery.

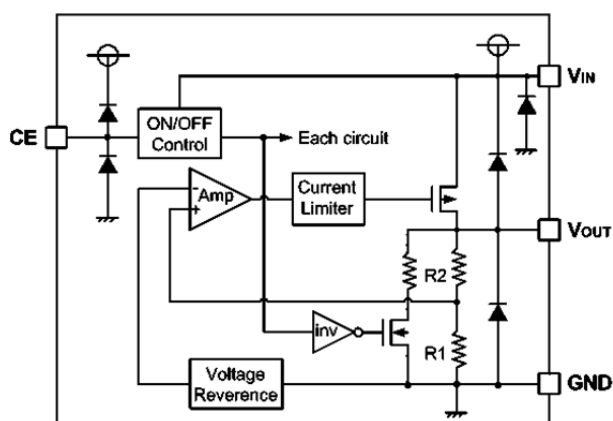
■ FEATURES

- Low Quiescent Current: 5 μ A
- Output Current: 300mA
- Output Voltage Range: 0.9V ~ 5.0V, (selectable in 0.1V steps)
- High Accuracy: $\pm 2\%$ (Typ.)
- Dropout Voltage: 150mV@100mA (3.0V Typ.)
- Excellent Line Regulation: 0.1%/V
- Built-in Current Limiter
- Built-in Short Circuit Protection
- Static safety: 2KV@HBM
- TC: 100ppm/ $^{\circ}$ C
- Low ESR Ceramic Capacitor Compatible

■ APPLICATIONS

- Battery powered systems
- Portable instrumentations
- Reference Voltage Sources
- Portable consumer equipments
- Portable/Palm computers
- Radio control systems

■ BLOCK DIAGRAM



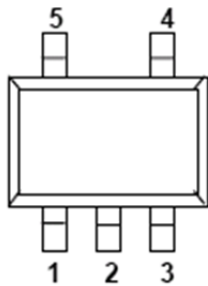
■ ORDER INFORMATION

CE6218①②③④

DESIGNATOR	SYMBOL	DESCRIPTION
①	E	With Shutdown Function
②③	Integer	Output Voltage (0.9~5V) e. g: 3.0V=②:3,③:0
④	M	Package: SOT-23-5
	P	Package: SOT-89-5

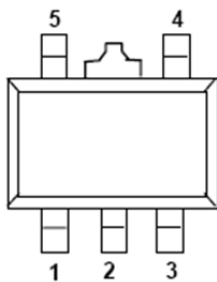
■ PIN CONFIGURATION (Pin output sequence can be ordered by customer)

SOT-23-5



PIN NUMBER	SYMBOL	FUNCTION
1	V_{IN}	Power Input Pin
2	V_{SS}	Ground
3	CE	Chip Enable Pin
4	NC	No Connection
5	V_{OUT}	Output Pin

SOT-89-5



PIN NUMBER	SYMBOL	FUNCTION
1	V_{OUT}	Output Pin
2	V_{SS}	Ground
3	NC	No Connection
4	CE	Chip Enable Pin
5	V_{IN}	Power Input Pin

■ ABSOLUTE MAXIMUM RATINGS

PARAMETER		SYMBOL	RATINGS	UNITS
Input Voltage		V_{IN}	$V_{SS} - 0.3 \sim V_{SS} + 8$	V
Output Current		I_{OUT}	600	mA
Output Voltage		V_{OUT}	$V_{SS} - 0.3 \sim V_{IN} + 0.3$	V
Power Dissipation	SOT-23	P_d	400	mW
	SOT-89	P_d	600	mW
Operating Temperature		T_{opr}	$-40 \sim +85$	$^{\circ}\text{C}$
Storage Temperature		T_{stg}	$-40 \sim +125$	$^{\circ}\text{C}$
Soldering Temperature & Time		T_{solder}	$260^{\circ}\text{C}, 10\text{s}$	

■ ELECTRICAL CHARACTERISTICS

CE6218 Series

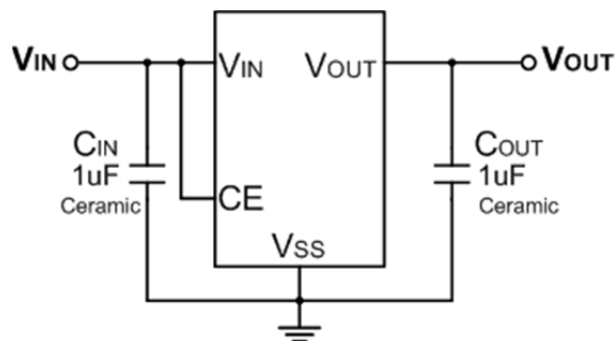
(C_{IN} = C_{OUT} = 1μF, Ta=25°C)

PARAMETER	SYMBOL	CONDITIONS		MIN	TYP	MAX	UNITS
Output Voltage	V _{OUT} (E) (Note 2)	I _{OUT} =40mA V _{IN} =V _{OUT} +1V	1.5V<V _{OUT} ≤5.0V	V _{OUT} *0.98	V _{OUT}	V _{OUT} *1.02	V
			0.9V≤V _{OUT} ≤1.5V	V _{OUT} -0.03	V _{OUT}	V _{OUT} +0.03	V
Supply Current	I _{SS}	V _{CE} =V _{IN} =V _{OUT} +1V			5	10	μA
Shutdown Current	I _{SHDN}	V _{CE} = V _{SS}				0.2	μA
Output Current	I _{OUT}	—		300			mA
Dropout Voltage (Note 3)	V _{diff1}	I _{OUT} = 40mA			60		mV
	V _{diff2}	I _{OUT} = 100mA			150		mV
Load Regulation	ΔV _{OUT}	V _{IN} = V _{OUT} +1V, 1mA≤I _{OUT} ≤100mA			10		mV
Line Regulation	$\frac{\Delta V_{OUT}}{\Delta V_{IN} * V_{OUT}}$	I _{OUT} = 40mA V _{OUT} +1V≤V _{IN} ≤6V			0.1	0.3	%/V
Output Voltage Temperature Characteristics	$\frac{\Delta V_{OUT}}{\Delta T * V_{OUT}}$	I _{OUT} = 40mA -40≤T≤+85			100		ppm/°C
Power Supply Ripple Rejection	PSRR	I _{OUT} =10mA f = 1kHz			60		dB
Short Current	I _{Short}	V _{OUT} = V _{SS}			30		mA
Current Limit	I _{Lim}	V _{IN} = V _{OUT} +1V			600		mA
Input Voltage	V _{IN}	—		2.0		6.0	V
CE "High" Voltage	V _{CE} "H"			1.0		V _{IN}	V
CE "Low" Voltage	V _{CE} "L"					0.3	V
C _{OUT} Auto-Discharge Resistance	R _{DISCHRG}	V _{IN} =5V, V _{OUT} =3.0V, V _{CE} =V _{SS}			100		Ω

NOTE:

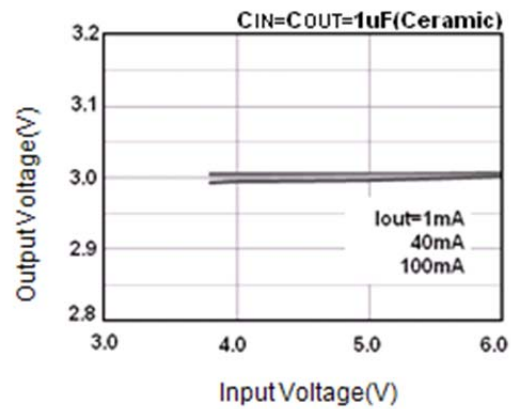
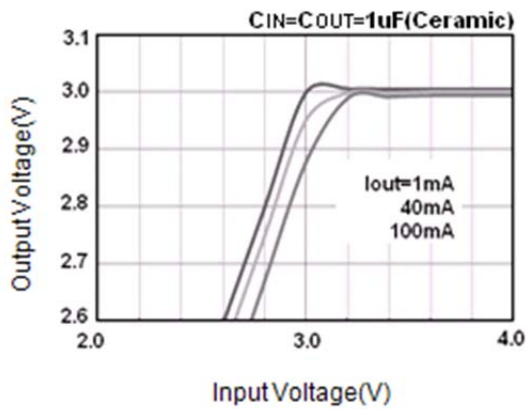
1. V_{OUT}: Specified Output Voltage.
2. V_{OUT} (E) : Effective Output Voltage (I.e. The Output Voltage When V_{IN} = (V_{OUT} +1.0V) And Maintain A Certain I_{OUT} Value).
3. V_{diff}: The Difference Of Output Voltage And Input Voltage When Input Voltage Is Decreased Gradually Till Output Voltage Equals To 98% Of V_{OUT} (E).

■ TYPICAL APPLICATION

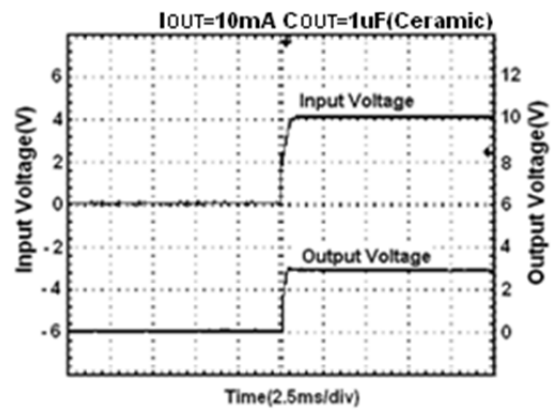
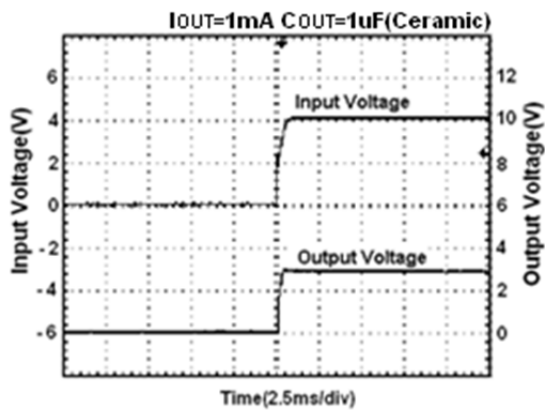


■ TYPICAL PERFORMANCE CHARACTERISTICS (CE6218E30M, for instance)

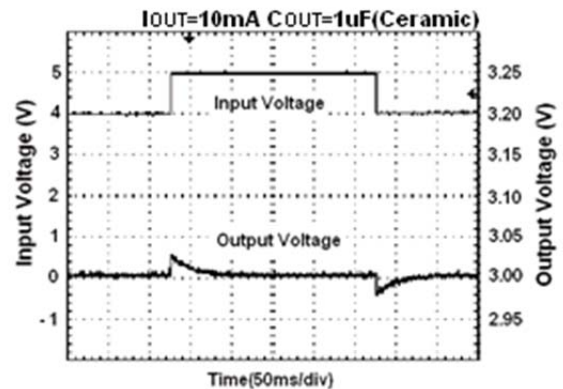
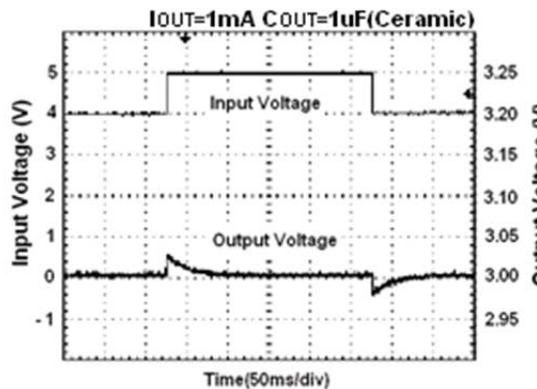
(1) Output Voltage vs. Input Voltage



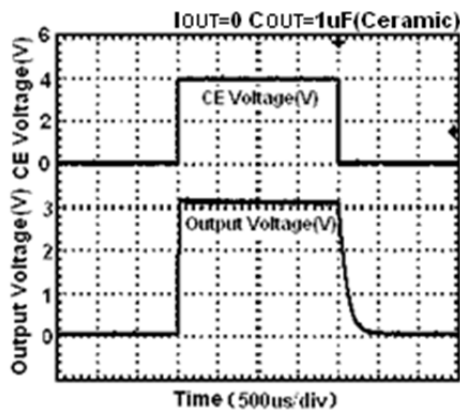
(2) Input Transient Response 1



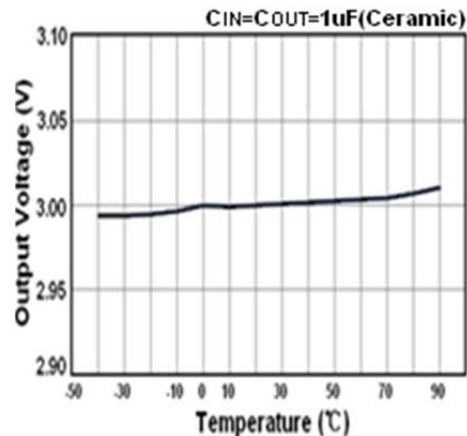
(3) Input Transient Response 2



(4) CE Shutdown Response



(5) Output Voltage vs. Temperature



C_{OUT} Auto-Discharge Function

CE6218 series can discharge the electric charge in the output capacitor (C_{OUT}), when a low signal to the CE pin, which enables a whole IC circuit turn off, is inputted via the N-channel transistor located between the V_{OUT} pin and the V_{SS} pin (cf. BLOCK DIAGRAM). The C_{OUT} auto-discharge resistance value is set at 100Ω ($V_{OUT}=3.0V$ @ $V_{IN}=5.0V$ at typical). The discharge time of the output capacitor (C_{OUT}) is set by the C_{OUT} auto-discharge resistance (R) and the output capacitor (C_{OUT}). By setting time constant of a C_{OUT} auto-discharge resistance value [$R_{DISCHRG}$] and an output capacitor value (C_{OUT}) as τ ($\tau=C \times R_{DISCHRG}$), the output voltage after discharge via the N-channel transistor is calculated by the following formulas.

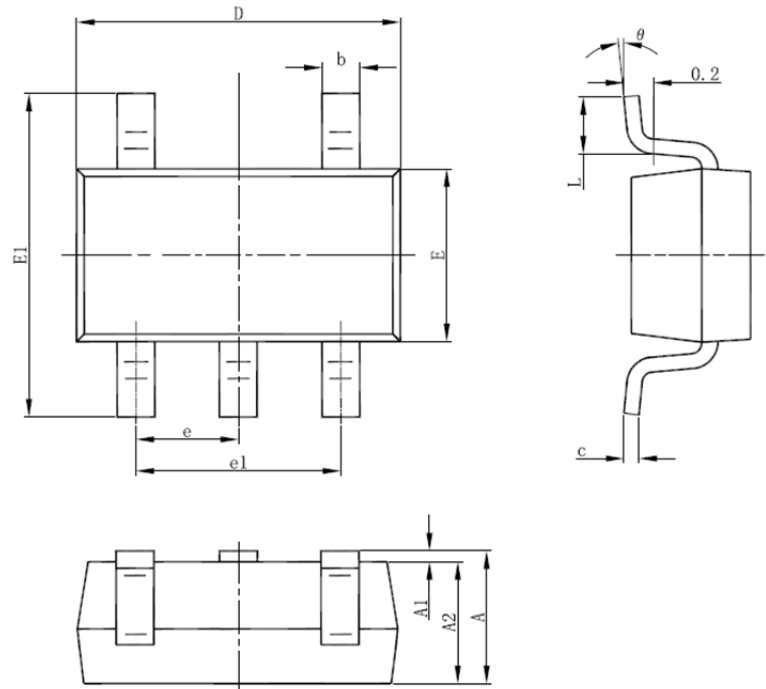
$$V = V_{OUT(E)} \times e^{-t/\tau}, \text{ or } t = \tau \ln(V / V_{OUT(E)})$$

(V : Output voltage after discharge, $V_{OUT(E)}$: Output voltage, t : Discharge time,

τ : C_{OUT} auto-discharge resistance $R_{DISCHRG}$ × Output capacitor (C_{OUT}) value C)

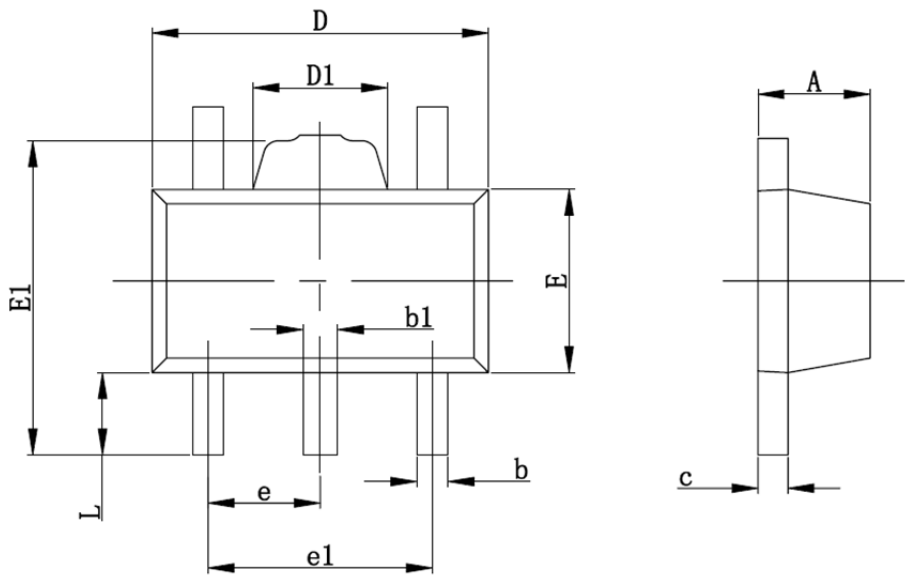
■ PACKAGING INFORMATION

● SOT-23-5 PACKAGE OUTLINE DIMENSIONS



Symbol	Dimensions In Millimeters		Dimensions In Inches	
	Min	Max	Min	Max
A	1.050	1.250	0.041	0.049
A1	0.000	0.100	0.000	0.004
A2	1.050	1.150	0.041	0.045
b	0.300	0.500	0.012	0.020
c	0.100	0.200	0.004	0.008
D	2.820	3.020	0.111	0.119
E	1.500	1.700	0.059	0.067
E1	2.650	2.950	0.104	0.116
e	0.950(BSC)		0.037(BSC)	
e1	1.800	2.000	0.071	0.079
L	0.300	0.600	0.012	0.024
θ	0°	8°	0°	8°

• SOT-89-5 PACKAGE OUTLINE DIMENSIONS



Symbol	Dimensions In Millimeters		Dimensions In Inches	
	Min	Max	Min	Max
A	1.400	1.600	0.055	0.063
b	0.320	0.520	0.013	0.020
b1	0.360	0.560	0.014	0.022
c	0.350	0.440	0.014	0.017
D	4.400	4.600	0.173	0.181
D1	1.400	1.800	0.055	0.071
E	2.300	2.600	0.091	0.102
E1	3.940	4.250	0.155	0.167
e	1.500TYP		0.060TYP	
e1	2.900	3.100	0.114	0.122
L	0.900	1.100	0.035	0.043

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